Yichao Jin, Ph.D.

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Health Economics | *Health Policy* | *Applied Econometrics* | *Causal Inference*

Education

University of Texas at Dallas May 2026
Ph.D. in Public Policy and Political Economy

- Dissertation: "Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan" — **ProQuest**
- Committee: Dohyeong Kim (Chair), Richard Scotch, Soyoung Kwon, Khai Chiong.
- Dean of Graduate Education Dissertation Research Award (2025).

University of Texas at Dallas May 2026
M.S. in Social Data Analytics and Research

University of Queensland 2019
Master of Development Economics

University of California, Riverside 2017
B.A. in Economics

Research Agenda

My research applies structural econometric methods to study how institutional design, regulatory incentives, and behavioral constraints shape health outcomes and policy effectiveness. I integrate discrete choice experiments, causal inference, and large-scale data analysis to evaluate how individuals and providers respond to policy changes. My work spans health behavior, health system performance, and the interaction between regulatory oversight and market outcomes.

Working Papers

1. **"Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust"**
   Under Review at Economic Analysis and Policy — **Link to Full Draft**
   - Structural econometric analysis of how institutional trust moderates the effect of time costs on health service utilization.
   - Identifies heterogeneous treatment effects relevant for targeting policy interventions across subpopulations.
2. **"Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework"**
   Working Paper (Target: J. Biomedical Informatics) — **SSRN Link**. DCE-grounded empirical-frontier regularization framework for LLM synthetic agents. Using a convex anchoring rule, reduces waiting-time monotonicity violations from 62.5% to 4.2% and improves Spearman rank correlation from −0.024 to 0.952. Presented at DAISY 2026 (ACM CHIL 2026) and AIxPH 2026.

3. **“Value Contamination in Policy Simulation”**
   SSRN Link. Audit framework measuring how model-inherent value orientations distort LLM-generated policy simulations.

4. **“The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount”**
   Manuscript in Preparation (Expected submission: August 2026) — Targeting *Social Science & Medicine*.

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.
- Leading development of the Behavioral Digital Twin (BDT) framework, integrating DCE data with causally-constrained LLM agents for synthetic policy simulation.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Conference Presentations

- **AIxPH 2026 (1st Annual Conference on AI and Public Health, Baltimore, MD)**: Empirical-Frontier Regularization for LLM Synthetic Agents — Poster Presentation.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026**: Behavioral Digital Twins — A Causally-Faithful Generative Framework.
- **UT Dallas Graduate Research Symposium 2025**: Modeling Vaccination Decisions with Hyperbolic Discounting.

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.
AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.
Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.
Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas
- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

REFERENCES

Dohyeong Kim (Chair)
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Robert E. Holmes Jr. Professor of Public Policy, GIS & Social Data Analytics
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